AP PRECALCULUS

Unit 1 • Worksheet 1.2A

Worksheet 1.2A • Rates of Change

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the average rate of change of $f(x) = 2x^2 - 3$ on $[1, 4]$?

(A) 10
(B) 15
(C) 30
(D) 8

Question 2

NO CALCULATOR

What is the average rate of change of $g(x) = x^3 - 2x$ on $[-1, 2]$?

(A) 3
(B) $\frac{5}{3}$
(C) 1
(D) -1

Question 3

NO CALCULATOR

What is the average rate of change of $h(x) = \frac{12}{x}$ on $[2, 6]$?

(A) -1
(B) -4
(C) 1
(D) $-\frac{1}{2}$

Question 4

NO CALCULATOR

The average rate of change of $f(x) = x^2 + kx$ on $[0, 2]$ is 5. What is k?

(A) 1
(B) 5
(C) 3
(D) 7

Question 5 (Free response) NO CALCULATOR

Let $f(x) = x^2 - 6x + 10$.

(a) Find the average rate of change of f on $[0, 3]$ and on $[3, 6]$.
(b) Explain what the signs of your answers tell you about f on each interval.
(c) Find a value $b > 0$ such that the average rate of change of f on $[0, b]$ is 0.